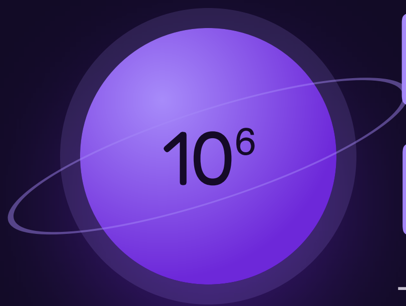
ADVENTURES IN BIG MATH · YEAR ONE

MISSION 03 OF 08 · WEEKS 11–14



Number Power

The mission where the rules turn out to be agreements.

THE BIG QUESTION

Why do we need rules about which operation goes first?

HANDS-ON PROJECT

A Million Dots

Estimate how much space a million of something takes. Then prove it, by counting one square and scaling up.

FLUENCY GAME

Expression Duel

Four cards, any operations and brackets. Build the biggest value you can defend.

AXIOM · MISSION COMMANDER, NINE TOURS

“Brock — this one is short and it's the one people skip. Skip it and every later mission gets harder for no reason.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

NAME ______________

DATE ______________

This is not a test. Nothing here counts for a grade. It exists to delete work you don't need. This is a four-week mission, so a strong showing here can shorten it to two.

STRAND A
READING NUMBERS

A1 Write 3,472,861 in words.

A2 Write six million forty thousand five in digits.

STRAND B
PLACE VALUE

B1 In 3,472,861 what is the 4 worth?

B2 The 6 in 6,000 is worth how many times the 6 in 60?

STRAND C
ROUNDING

C1 Round 47,382 to the nearest thousand.

C2 Round the same number to the nearest ten thousand.

STRAND D
ESTIMATION

D1 Estimate 412 × 19 without computing it.

D2 Is 38 × 52 nearer 200, 2,000 or 20,000? How do you know?

STRAND E
ORDER OF OPS

E1 $5 + 3 \times 4 =$ _______

E2 $18 \div (3 + 6) =$ _______

STRAND F
EXPONENTS

F1 $2^3 =$ _______ $5^2 =$ _______

F2 How many zeros does 10^6 have? Why that many?

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Reading numbers	A1 A2	Week 1 · lesson 1.1	___ / 2	Day 1.1. Warm-Up tier drops to 3 items all unit.
B · Place value	B1 B2	Week 1 · lesson 1.2	___ / 2	Day 1.2 — but keep the Million Dots project either way.
C · Rounding	C1 C2	Week 1 · 1.3 and Week 2 · 2.1	___ / 2	Day 1.3 and day 2.1.
D · Estimation	D1 D2	Week 2	___ / 2	Days 2.1–2.2. Never skip 2.3, the Fermi questions.
E · Order of ops	E1 E2	Week 1 · 1.4 and Week 3	___ / 2	Days 3.1–3.2. Keep 3.4 — writing expressions is the hard part, not reading them.
F · Exponents	F1 F2	Week 4 · lessons 4.1–4.2	___ / 2	Days 4.1–4.2. The capstone display still happens.

Where the saved time goes

This mission is already short. Compacted time goes into A Million Dots — the display is worth more than another worksheet — or straight on to Mission 04. Write the reallocation here:

....

....

Revised Mission 03 plan

WEEK 1 ☐ full ☐ compressed ☐ skip

WEEK 2 ☐ full ☐ compressed ☐ skip

WEEK 3 ☐ full ☐ compressed ☐ skip

WEEK 4 ☐ full ☐ compressed ☐ skip

NAME ___________

DATE ___________

How Big Is a Million

Seven places, each one ten of the place on its right.

Warm-Up

6 ITEMS · 5 MINUTES

tens in 100	hundreds in 1,000	thousands in 10,000	10 × 100	100 × 100	1,000 × 10
___	___	___	___	___	___

THE CHART · FILL IN THE VALUE OF EACH DIGIT

3	4	7	2	8	6	1
MILLIONS	HUND THOUS	TEN THOUS	THOUSANDS	HUNDREDS	TENS	ONES
....	....	....	....	....	....	....

Core

SAY THE PLACE OUT LOUD BEFORE YOU WRITE THE VALUE

1. In 3,472,861, what is the **4** worth? place ________ ....

2. In 3,472,861, what is the **7** worth? place ________ ....

3. In 5,208,043, what is the **2** worth? place ________ ....

4. How many thousands make 4,000,000? ....

5. Write *six million forty thousand five* in digits. ....

Challenge

NOT GRADED

C1. How many zeros are in one million? Write it out and count them.

....

C2. 1,000 × 1,000 = _______. Explain why the zeros add up the way they do.

....

C3. If you counted one number per second, roughly how long would counting to a million take? Guess first, then work it out.

....

NAME ___________

DATE ___________

Ten Times Bigger

Multiplying by ten shifts every digit one place left. The zeros are the evidence, not the rule.

Warm-Up

40 × 10	700 × 10	60 × 100	8 × 1,000	3,200 ÷ 10	45,000 ÷ 1,000
____	____	____	____	____	____

THE MOVE

Every digit slides one place left.

230 × 100 → 23,000

Two zeros arrived because the 2 and the 3 each moved two places, and the empty places had to be held.

Core

1. 230 × 100 ....

2. 9,000 ÷ 100 ....

3. 56 × 1,000 ....

4. 740,000 ÷ 10,000 ....

5. How many times bigger is 800,000 than 800? Say how you know. ....

Challenge

C1. The 6 in 6,000 is worth how many times the 6 in 60? Show the places you counted.

....

C2. How many 10,000s make one million?

....

C3. Someone says “multiplying by 10 just means adding a zero.” Find a case where that sentence gets you into trouble.

....

NAME ___________

DATE ___________

Compare, Order, Round

Rounding answers the question you were asked, not the number you were given.

Warm-Up

ROUND TO THE NEAREST TEN

47	83	65	128	254	999
____	____	____	____	____	____

WORKED EXAMPLE · 47,382

To the nearest thousand

382 down	618 up

Down is the shorter trip → **47,000**

To the nearest ten thousand

7,382 down	2,618 up

Now up is shorter → **50,000**

Core

1. Round 47,382 to the nearest **thousand**. ....
2. Round 47,382 to the nearest **ten thousand**. ....
3. Round 863,209 to the nearest **hundred thousand**. ....
4. Which is larger, 408,916 or 480,169? Write < or > between them. ....
5. Round 2,499 to the nearest **hundred**. ....

Challenge

C1. What is the largest number that rounds to 5,000 at the nearest thousand?

....

C2. And the smallest? Explain what happens exactly at the halfway point.

....

C3. Find a number that rounds to 50,000 at the ten thousand but to 47,000 at the thousand.

....

NAME ___________

DATE ___________

Order of Operations

Build the groups first. Add and subtract what's left loose. Brackets overrule everything.

Warm-Up

$3 + 4 \times 2$	$10 - 2 \times 3$	$12 \div 4 + 5$	$6 + 6 \div 2$	$2 \times 3 + 4 \times 5$	$20 - (4 + 6)$
____	____	____	____	____	____

$5 + 3 \times 4$

3×4
12

$+ 5$
loose

$12 + 5 = 17$

The 5 was never part of the group, so it can't join in until the group is built. Brackets are how you argue with that.

Core

WRITE THE STEP YOU DID FIRST, THEN THE ANSWER

1. $5 + 3 \times 4$ ....
2. $(5 + 3) \times 4$ ....
3. $18 \div (3 + 6)$ ....
4. $40 - 6 \times 5 + 2$ ....
5. $3 \times (12 - 4) \div 6$ ....

Challenge

C1. $2 + 3 \times 4 - 6 \div 2 =$ ______. Write every step in order.

....

C2. $(8 + 4) \times (9 - 6) =$ ______

C3. Put brackets into $2 + 4 \times 5 - 1$ so that it equals 29. Then find a placement that gives 21.

....

NAME ___________

DATE ___________

A Million Dots Begins

Guess first. Write the guess down before you measure anything — that's the whole discipline.

The project · four steps

1 · Guess. How big a sheet would hold a million dots? Write the guess now.

2 · Count one. Draw dots in one square centimetre. Count them exactly.

3 · Scale up. Multiply. How many square centimetres do you need?

4 · Compare. Put your answer next to your guess. Which was bigger, and by how much?

My guess: ....

Powers

THE LITTLE NUMBER COUNTS COPIES, NOT MULTIPLES

2^3	5^2	10^4	$3^2 + 4^2$ ....
$2^5 - 2^3$	$10^6 \div 10^3$	Bigger: 4^3 or 3^4?	Zeros in 10^8? ....

Expression Duel

Deal four number cards. Each player builds one expression using all four cards and any operations, brackets and powers. Highest value wins the round — but you must be able to evaluate your own expression correctly, or it scores zero.

My cards: ....

My expression: ....

Its value: ....

A better one I found after: ....

The real question

With four cards, is it always best to use powers? Find a set of four cards where a power gives you the biggest value, and another set where plain multiplication beats it.

....

....

The Rest of the Mission

Three weeks left, not four — this is the short mission. Same three-tier shape every day.

Week 2 · Estimation

ROUNDING BECOMES A HABIT, NOT A TASK

2.1 Mon
Estimate first, always. Round both numbers, then compute for real.

2.2 Tue
How wrong was it? High or low, and why.

2.3 Wed
Fermi questions. Estimate with reasons instead of facts.

2.4 Thu
Is that answer sensible? Pick the only possible one without computing.

Fri
A Million Dots, measured. Count one square, scale it up.

Week 3 · Order of Operations

MID-UNIT QUIZ FRIDAY · THE BIG QUESTION WEEK

3.1 Mon
Two answers, one expression. Why a rule was necessary at all.

3.2 Tue
Groups go first — and the rectangle shows why.

3.3 Wed
Brackets overrule. Same numbers, deliberately different answer.

3.4 Thu
Write your own: build 24 three different ways.

Fri
Mid-unit quiz, 8 items, Weeks 1–3.

Week 4 · Exponents & Capstone

UNIT TEST FRIDAY · BADGE AWARDED

4.1 Mon
The little raised number. 2^3 is not 2×3.

4.2 Tue
Powers of ten. Place value and exponents turn out to be one idea.

4.3 Wed
Expression Duel with powers allowed.

Thu
Million Dots display finished; error journal sweep.

Fri
Mission 03 test: 12 items + one explanation.

MATERIALS, WHOLE UNIT

Graph paper (¼ in) · a metre of butcher paper for the display · fine-tip pens · deck of cards · calculator, for checking only · the error journal

WEEKLY CHECK

Five items every Friday, mixed tiers. Below 4/5 means Monday is a reteach, not new material.

THE MOST COMMON STALL

Reading 2^3 as 6. It's not carelessness — it's a genuine misread. Draw the cube once and it usually sticks.

Mid-Unit Quiz

8 items · Weeks 1–3 · 7 of 8 keeps you flying

NAME ___________

DATE ___________

___ / 8

1. In 3,472,861, what is the 4 worth?

2. Write five million two hundred thousand nine in digits.

3. 230 × 100 = ______ and 9,000 ÷ 100 = ______

4. Round 863,209 to the nearest hundred thousand.

5. Estimate 412 × 19 by rounding both numbers. Write the estimate, then say whether the real answer will be higher or lower than it.

6. 40 − 6 × 5 + 2 = _______

7. 3 × (12 − 4) ÷ 6 = _______

CHALLENGE ITEM

8. 3 + 4 × ☐ = 23 ☐ = _____

What you worked out first:

NAME ___________

DATE ___________

Mission 03 Test

12 items + one explanation · part 1 of 2

1. In 5,208,043, what is the 2 worth?

2. Write 4,060,900 in words.

3. $56 \times 1{,}000 =$ ______

4. $740{,}000 \div 10{,}000 =$ ______

5. Round 47,382 to the nearest thousand, then to the nearest ten thousand.

6. Put < or > between 408,916 and 480,169.

7. Estimate 587 × 41. Then say whether your estimate is above or below the real answer, and why.

8. $5 + 3 \times 4$

9. $18 \div (3 + 6)$

10. $(8 + 4) \times (9 - 6)$

Reasoning & the Big Question

ITEMS ___ / 12

EXPLANATION ___ / 4

11. 2^3 = _______ 5^2 = _______ 10^4 = _______

12. Someone writes $2^3 = 6$. What did they misread, and what is the right answer?

CHALLENGE ITEM

Put brackets into $2 + 4 \times 5 - 1$ so that it equals 29, then so that it equals 21.

EXPLAIN YOUR THINKING · WORTH 4 POINTS

Why do we need rules about which operation goes first?

Write one expression that two people could read two different ways. Show both answers. Then say which one the rules pick, and draw a picture showing why that reading is the sensible one. Spelling doesn't count. Reasoning does.

TROPHY BAND · CIRCLE ONE

85–100%

Mission Commander

Powers Badge earned. Mission 04 opens — and Challenge becomes your Core.

70–84%

Flight Engineer

Two targeted days on the flagged items, then retake those items only.

Below 70%

Ground Crew

Rebuild the place-value chart by hand before going near exponents again.

Key A · Pre-Assessment, Lessons 1.1–1.2

Pre-Assessment

A1 three million, four hundred seventy-two thousand, eight hundred sixty-one

A2 **6,040,005**

B1 **400,000**

B2 **100 times**

C1 **47,000**

C2 **50,000**

D1 ≈ 400 × 20 = **8,000**

D2 **2,000** — 40 × 50 = 2,000

E1 **17**

E2 **2**

F1 **8** and **25**

F2 **six zeros** — one per ten multiplied

Scoring note. A1 scores for correct place names, not spelling. E1 answered as 32 is the classic misread — mark the strand new and run Week 3 in full.

Lesson 1.1 · How Big Is a Million

Warm-Up 10 · 10 · 10 · 1,000 · 10,000 · 10,000

Chart 3,000,000 · 400,000 · 70,000 · 2,000 · 800 · 60 · 1

Core

1. hundred thousands · **400,000**
2. ten thousands · **70,000**
3. hundred thousands · **200,000**
4. **4,000** thousands
5. **6,040,005**

Challenge. C1 · six zeros. C2 · 1,000,000; each factor contributes three zeros, so the product has six. C3 · about 11½ days of non-stop counting (1,000,000 seconds ÷ 86,400). Accept any answer near "a bit over a week" with the working shown.

Lesson 1.2 · Ten Times Bigger

Warm-Up 400 · 7,000 · 6,000 · 8,000 · 320 · 45

1. **23,000** 2. **90** 3. **56,000**

4. **74** 5. **1,000 times** (three places)

Challenge. C1 · 100 times — two places apart. C2 · 100. C3 · the sentence breaks on decimals: 2.5 × 10 is 25, not 2.50. Accept any counterexample, or a clear statement that digits move places and zeros only hold the empty ones.

Key B · Lessons 1.3–1.4, Enrichment, Quiz

Lesson 1.3 · Compare, Order, Round

Warm-Up 50 · 80 · 70 · 130 · 250 · 1,000

Core

1. **47,000**
2. **50,000**
3. **900,000**
4. 408,916 < 480,169
5. **2,500**

Challenge. C1 · 5,499. C2 · 4,500 — exactly halfway rounds up by convention, and it is a convention, not a law; say so if he asks. C3 · any number from 47,000 to 47,499, e.g. 47,382 itself.

Lesson 1.4 · Order of Operations

Warm-Up 11 · 4 · 8 · 9 · 26 · 10

1. 3 × 4 first → **17**
2. brackets first → **32**
3. 3 + 6 = 9 → **2**
4. 6 × 5 = 30 → **12**
5. 12 - 4 = 8, 3 × 8 = 24 → **4**

Challenge. C1 · 2 + 12 – 3 = **11**. C2 · **36**. C3 · (2 + 4) × 5 – 1 = 29; 2 + 4 × (5 – 1) = 18, so for 21 use (2 + 4 × 5) – 1 = 21. Accept any correct placement — and the discovery that some targets are impossible is itself a good answer.

Friday · A Million Dots & Powers

Powers 2^3 = **8** · 5^2 = **25** · 10^4 = **10,000**

$3^2 + 4^2$ = **25** · $2^5 - 2^3$ = **24**

$10^6 \div 10^3$ = **1,000** · zeros in 10^8 = **8**

Bigger: 3^4 = **81** beats 4^3 = 64

The project. There is no single right answer — the mark is for the method. A typical result: about 25 dots fit comfortably in one square centimetre, so a million needs roughly 40,000 cm², which is a square about 2 metres on a side. Anything between one and four metres, with the working shown, is excellent.

The real question. Powers win when the cards include a small base and a large exponent that isn't 1 — 2 and 9 give 512, far past 2 × 9. Plain multiplication wins when the numbers are close and large: 8 × 9 = 72 beats 8^9 being unavailable with only those cards, and beats 9^2 = 81 only if a third card is in play. Accept any pair of examples with the arithmetic done correctly.

Mid-Unit Quiz · out of 8

1. **400,000**
2. **5,200,009**
3. **23,000** and **90**
4. **900,000**
5. ≈ 400 × 20 = **8,000**; real answer 7,828, so the estimate is high
6. **12**
7. **4**
8. ▢ = **5**

7/8 or better → continue. A miss on 6 or 7 is the order of operations, not arithmetic — reteach with the rectangle picture, not with more practice.

Key C · Mission 03 Test & Scoring

Items 1–12 · one point each

1. **200,000**
2. four million, sixty thousand, nine hundred
3. **56,000**
4. **74**
5. **47,000** then **50,000**
6. 408,916 **<** 480,169
7. ≈ 600 × 40 = **24,000**; real 24,067, so the estimate is slightly low
8. **17**
9. **2**
10. **36**
11. **8 · 25 · 10,000**
12. They read it as 2 × 3. It means 2 × 2 × 2 = 8.

Explanation · 4 points

Score the reasoning, not the writing. Award one point each:

- ☐ 1 · Writes an expression with two plausible readings
- ☐ 2 · Computes both answers correctly
- ☐ 3 · Says which one the rules pick
- ☐ 4 · Explains why — a group exists before the loose numbers join it

Model answer, kid-sized. "5 + 3 × 4 could be 32 or 17. The rule says 17, because 3 × 4 is already a group of twelve and the 5 is just sitting next to it. If everyone did it their own way we'd all get different answers off the same line."

Total 16 points. 14+ Mission Commander · 11–13 Flight Engineer · 10 or fewer Ground Crew. Item 12 is diagnostic — a miss there means Week 4 gets one more day before Mission 04.

Error Journal · photocopy this page

One row per mistake, for the whole mission. The "why" line must name a step. "I was careless" doesn't count.

DATE	PROBLEM	I GOT / ANSWER	WHY IT HAPPENED — ONE SENTENCE